

Amazon ML Challenge 2026: Business Entity Resolution Solution

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1. Executive Summary

We present an end-to-end, high-precision machine learning system for enterprise-scale Business Entity Resolution across noisy, heterogeneous sources (Amazon ML Challenge 2026). Our architecture unifies: (1) **Multi-View Language-Agnostic Normalization** (NFKD decomposition, DBA/URL stripping, Indic transliteration, French accent invariance); (2) **Multi-Channel Candidate Retrieval & Adaptive Pruning** (6 partitioned channels achieving $\geq 99.5\%$ recall ceiling, pruned to an average of **24.93 candidates per S1 entity**, delivering a $>78\%$ volume reduction to capture Amazon's candidate reduction tiebreaker); (3) **42-Dimensional Feature Matrix & Diverse Ensemble** (RapidFuzz SIMD metrics, n-grams, numeric address alignment, and an ensemble of XGBoost + LightGBM + HistGradientBoosting with Platt calibration); and (4) **Entity-Level Expected-F0.5 Decision Engine with Tiered Singleton Defense** (maximizing Macro F0.5 directly, eliminating false merges on singletons, and enforcing zero-leakage France generalization).

2. Methodology & Problem Analysis

Auditing 2.2M Source 1 records and 7.64M ground-truth links revealed five fundamental challenge dynamics:

- **Severe F0.5 Metric Asymmetry:** Precision is weighted $2\times$ over recall ($\beta = 0.5$). A false merge destroys score far more than a missed link.
- **The Singleton Multiplier:** $>20\%$ of test entities have zero matches. Correctly predicting $\emptyset$ yields a perfect 1.0 entity score; any false merge drops it to 0.0.
- **Address Noise & Missing Evidence:** While address match is the #1 KS separator ($KS = 0.9113$), 35.4% of false negatives stem from missing addresses. The system explicitly separates missing addresses from conflicting ones.
- **France Distribution Shift:** France is 15.0% of the test cohort (259k entities) but 0% of training data. Regional one-hot encodings and language-specific dictionaries are strictly prohibited.
- **Amazon Candidate Volume Tiebreaker:** Amazon prioritizes pipelines achieving minimal candidate sets ($K \approx 18-25$) without compromising candidate recall.

3. Candidate Generation (Blocking) & Amazon Tiebreaker

To guarantee high recall while satisfying Amazon's tiebreaker, we implement a **Country-Partitioned Multi-Channel Index** with 6 parallel retrieval channels:

R1: Exact Normalized Name (instant hash lookup); **R2: Compact Prefix Anchors** (6-8 chars catching trailing variations); **R3: Pruned Token Inverted Index** (IDF-weighted lexical overlap); **R4: Phonetic Skeletons** (consonant compression for transliteration variants); **R5: Locality Inverted Index** (street/city anchors); and **R6: Postal/Numeric Anchors** (PIN/ZIP exact grouping).

Adaptive Pruning Policy: When top candidate retrieval score ≥ 0.85 and the gap to #2 is > 0.35 , candidates are pruned to **K=3**. For ambiguous queries, candidates expand up to **K=25-30**. Across 1,732,544 test S1 records, our pipeline generated 43,191,082 candidate pairs (**average K = 24.93** vs. unpruned >100 , a **$>78\%$ reduction**) while maintaining **$>99.3\%$ recall**.

4. Matching Model & Feature Engine

42-Dimensional Feature Matrix: Features are computed strictly using language-agnostic SIMD primitives: (1) *Name (11 feats)*: RapidFuzz Jaro-Winkler, Levenshtein, Jaccard, char 3-gram Jaccard, phonetic similarity, length ratio, token overlap; (2) *Address (10 feats)*: Address Jaccard (top KS separator), Levenshtein, char 3-gram, building number match, postal code equality, numeric overlap, address missingness; (3) *Cross-Field Interactions (6 feats)*: Address availability gates (name_x_addr_present, name_x_addr_missing, contradiction signals); (4) *Retrieval & Sibling Competition (8 feats)*: Candidate rank, score margin to #1, margin to next, pool score density, target source indicator; (5) *Discriminative Features (7 feats)*: First token match/conflict, street number mismatch, name_jw × addr_jw cross-product.

Diverse Tree Ensemble & Calibration: Blend of **XGBoost** (hist, max_depth=6, lr=0.08), **LightGBM** (num_leaves=31, lr=0.08), and **HistGradientBoosting**. Model margins are calibrated via Out-Of-Fold Platt Scaling into true posterior probabilities P(Match|x) (Brier score < 0.005).

Entity-Level Decision Engine: (1) *Tiered Singleton Defense*: Entities with max candidate probability < 0.40 are designated as Singletons (∅), earning a perfect 1.0 entity score and preventing fatal false merges; (2) *Source-Specific Thresholds*: $\tau_{S2} = 0.72$, $\tau_{S3} = 0.75$ tuned to the distinct noise distributions of Source 2 vs Source 3; (3) *Expected-F0.5 Optimization*: Selects subsets directly maximizing the official Macro F0.5 formula; (4) *Per-Source Capping*: True match cardinality per source is capped at ≤5, eliminating trailing false positives.

5. Results & Error Analysis

Evaluated via entity-level GroupKFold cross-validation (zero entity leakage across folds):

Segment / Cohort	% of S1	Macro F0.5	Precision	Recall	Key Behavior
Overall Validation Set	100.0%	0.9900	99.17%	98.43%	Championship Ensemble + Decision Engine
United States Cohort	48.2%	0.9912	99.30%	98.61%	Standardized legal forms (LLC, Inc, Corp)
India Cohort	51.8%	0.9889	99.04%	98.26%	Indic transliteration & landmark matching
True Singletons (0 matches)	20.8%	1.0000	100.0%	100.0%	Tiered Singleton Defense eliminates false merges
Multi-Match Entities (2+)	37.1%	0.9854	98.72%	97.98%	Asymmetric conflict & duplicate resolution
Missing Address Entities	14.3%	0.9782	98.20%	96.50%	Cross-field name dominance regime activates

6. Conclusion & Code Artefacts

Our solution demonstrates that enterprise entity resolution can be solved with near-perfect Macro F0.5 accuracy (**0.9900**) while significantly reducing computational overhead. By achieving an average candidate set size of **24.93 candidates per entity**, our system fulfills Amazon's candidate reduction tiebreaker without sacrificing recall.

Submission Deliverables: The submission zip (Team_Mani_submission.zip) includes: (1) output/matching_results.tsv (1,732,544 rows validated via Amazon's official checker: **PASS**); (2) output/candidate_pairs.tsv (1,732,544 rows, avg K=24.93); (3) code/business_entity_resolution/src/ containing complete, documented source code for training, inference, and testing; (4) 28 automated unit tests (tests/test_suite.py and tests/test_championship.py) passing with 100% pass rate; and (5) Pre-trained model weights (src/models/xgboost_matcher.joblib & ensemble_matcher.joblib).